

Hi, and welcome to the Human AI Interaction course.

This will be a highly interactive experience where you'll learn how to envision, design and test interactive systems, with a specific focus on how AI, in the many shapes and forms that it now takes, can serve as a design material, as something that you can work with and design and make available to people.

This is a new course, and one that's based around design thinking, so I will do my best really shape this around the interests and needs of those of you who end up taking this class. In our previous bachelor's module,

Human-Computer Interaction, we've had students go off to the industry as designers or UX researchers, as well as continue with masters and PhDs in the area of Human- Computer Interaction.

So whether you're just browsing through during the induction week or you've already committed to taking this, I really hope that you'll find the material interesting and also potentially directly applicable to your next career steps.

Timestamp 1 minute I'm Peter Slovak and I'll be your lecturer this term together with Dariana who's my PhD student and who will be working with you during the small group tutorials which are straight after the two-hour lectures so I hope that most of the times I might be

able to join in for those as well. Just to give you a sense about who I am, I'm a reader in Human-Computer Interaction here at King's College London at the Informatics department but I've also been fortunate to have a couple of additional visiting positions that kind of fall out of my own research and a fellowship that I hold, including one at the Child and Adults in Psychiatry here at King's, where most of my group is currently based, as well as a couple of visiting position hats at University of Oxford.

In case anyone's interested in the type of work that

I do for my research, it's really close to what we will be talking about here, or at least some aspects of what we'll be talking about in the course, it's really looking at how would you use generative AI to develop systems that can support mental health or well-being.

And I just put a link to a talk that I gave just a couple of weeks ago for you to potentially have a look if you want to. It's obviously totally optional.

The ends of this course are mainly to give you both the theory as well as practical tools so that you can envision, design, prototype, and evaluate interactive systems that are supposed to help humans do whatever they want to do. Most of the material will be grounded in talking about how to

design desktop and mobile apps, but we'll also touch on some of the more novel and emerging interaction modes, such as voice-based interaction.

All the material will be infused with a tidal wave of generative AI that is now emerging over the last couple of years, and which promises an entirely new way of engaging with technology as well as approaches to design.

As you can imagine, the speed and novelty is both a blessing and a curse. It is an absolutely exciting Timestamp 3 minutes time within HCI where many of the approaches are being re-envisioned nearly literally as we speak.

But it's also somewhat frustrating and confusing as there yet to emerge any hard-wired rules about what are the right ways to design with this technology.

This is the reason for us to focus the material in this course in three main areas. The first is what you might call a traditional interaction design.

These might be the rules and design principles that we will be underpinning most of the apps and websites that you might have seen perhaps three to four years ago, and that has a long history for HCI at least, which goes back for perhaps 10-15 years. The second area is then thinking about specifically what does it mean to design with traditional deep learning,

where you as a designer might be working with machine learning approaches and algorithms for which you've selected most of the data for or where your team has selected most of the data and where you're still doing most of the model training as an organization.

So that's where the questions of bias and Timestamp 4 minutes fairness and trust are really crucial.

Finally we'll be exploring the emerging new world of generative AI and the design with foundational models where you're working with these big entities that have cost ridiculous amounts of money and that are in that case outside of your control. You might fine-tune, but most of the time you're unable to really engage deeply with what is happening within the foundation model, and where we as a field are still trying to find out what to do and how to design with these.

Now, human AI interaction and interaction design more broadly is a bit like programming. There are some useful theories you should know, but what mostly matters is really learning by doing and being able to apply whatever the basic theory is in practice.

Part of this is that human-AI interaction involves humans, and AI, that's uncertain and stochastic, so it's pretty hard to figure out some unbreakable rules of how humans and AI operate in these ways.

So what you should expect from this course is that
Timestamp 5 minutes you'll be actively working with the concepts, and that

I expect that you'll be able to apply these into new situations, rather than just do a bit of rote learning or learn a couple of proofs and then repeat whatever was said exactly in the lecture during the exam.

I don't know if you've already seen this Bloom's taxonomy of learning, but my aim is that most of the time we'll be operating on these top four layers.

I'll assume that you remember and understand as a matter of principle, and we'll be focusing mostly on helping you understand how to apply and work with the concepts that we'll be talking about.

Given that this is going to be a very fast whistle-stop tour of HCI, where we essentially have one semester and one module to go all over all of the basics that might normally be done in more depth and across several courses in a more focused bachelor program will be relying on quite simple design thinking model from

Stanford. I personally quite like it and I think it does a great job at helping people get to grips with the ideas of what does it mean to design and how to

Timestamp 6 minutes get started with the design process. While it might not encompass all of the approaches and methods that you would want to do as an HCI researcher It seems to

also be closest to what would be happening in commercial practice and at least the best thing we could do when we're trying to fit into these 10 or 11 weeks.

What this means is that we'll be going through each of these phases in turn, introducing the theory as well as range of tools that you could use to deliver each step.

Again, the focus will be on the different aspects, if you can remember, basic interaction design, deep learning, as well as the generative AI learning.

And we will also be helping you immediately apply this on your coursework in your teams and get feedback on this as part of the weekly small group tutorials.

So let's now move on to the first actual concept and content of the course.

We'll start by talking about the core idea that will Timestamp 7 minutes be following us throughout the semester, the notion of interaction design.

So in the next 10 minutes or so, I want to walk you through the definitions as well as, perhaps more importantly, the examples of what interaction design means, where it's useful, and what kind of proceeds are useful to guide us.

As a reminder, we've already talked about the range of HCI areas and types of applications that interaction design encompasses, but it is actually even broader than that.

Feel free to stop the video if you want to, have a detailed look, as I won't bore everyone by stepping through this one by one.

However, I do want to point out two observations that might not be immediately apparent, especially if we're coming into what is interaction design from the viewpoint of, you know, let's go and design an app.

So the first one is that interaction design is actually bordering and overlapping with many areas that we wouldn't have thought of, including data visualization or architecture or industrial design. All of these have some aspects of designing interactions, whether that is the ways in which people walk through buildings

Timestamp 8 minutes or where they meet and how they meet, or whether it is

about how does the visualization become understandable and how does it make sense and how do people interact with it. The second key

point that I'd like to highlight and that we'll keep on coming back to as well, is how all of interaction design is subsumed by the broader notion of user experience. Which kind of makes sense, as designing interactions is just one part of thinking about what does a user experience look like.

We can have similar maps that look at AI and human-centred AI too. I've just put in this figure from a recent paper in one of the leading conferences in the field that really looks at the

different types and ways and approaches that we might take to thinking about AI.

Again, have a look at this, we might come back to this image later in the course. I just wanted to show at this moment how broad the area is and also how focused we will have to be as we step through it.

But really, interaction design can be seen as anything that involves technology of any certain people.

So that might include thinking about the interaction of how we manage a flow of people at the border, what do you ask them to do, where are they supposed to stand, how do they interact with the technology and other humans, or the ways in which every single piece of a car dashboard is designed to really optimize the interaction with the driver. Our lecture next week will get into some of the design decisions and theories that underpin such approaches. But let's start with a definition.

According to our textbooks, the interaction design is the design of interactive products to support the way people communicate and interact in their everyday and working lives.

Now that sounds really, really broad, but kind of makes sense if what we're trying to do is define something, a field, that can span both how do we design a border control to core dashboards, to apps, to everything

else. So now let's have a look at a few examples of what a good and bad designs look like, to get a better sense of what this breadth can actually mean.

So, this is an example of a bad interaction design, and it's probably bad because if I say you're driving past this, and it's Wednesday 4pm, and you're trying to park, are you allowed to park? Are you not allowed to park? No one knows. Well, you might know after you look at this for a long time, but that's what makes it a bad interaction. Now, let's have a look at the same information that is presented in a slightly different way.

Now, this could be considered a good interaction design because if you're thinking about Wednesday 4pm, well, it clearly is exactly the first time when you're allowed to park that day after 7am.

Both of these pieces are trying to communicate and enable people to interact with their environments, but they're doing it in different ways.

And part of what we're going to be learning in this course is how do you get from a bad one to a good one, and how do you recognize which one's which as you engage with it.

So let's keep on going with the definitions.

If the definition is what we've already said, what are the goals of interaction design?

As you could see from the previous example, the goals could be that it's something that's easy to learn, that it's something that's effective to use, and that ideally provides an enjoyable user experience, or at least one that is not entirely and utterly dreadful.

So, let's just have a quick look back at this image that we've had before.

The interaction design on the left side is neither easy nor effective, nor does it provide an enjoyable experience, unless you're the person going around and handing out tickets to people.

The one on the right might not be enjoyable, but at least it's something that visualizes and simplifies the information well enough so that people

Timestamp 12 minutes can do what they want to do, which in this case is to decide whether they can park at a given time.

So if we try to move past the definition, what could be the characteristics of what does it mean to do interaction design? The first characteristic is that design involves changing situations by shaping or deploying artifacts.

So the little things that show when and how you're supposed to park, that is an example of an artifact. So we're changing the situation that someone sees by,

for example, moving from left design to the right one.

It's about exploring possible futures, so often there's a problem, there's a terrifyingly bad version of showing someone when they're allowed to park, and thinking about what else could we do about this bad future and turn it into a possible future, which is hopefully not as bad.

It's about framing the problem in parallel with creating possible solutions.

So framing the problem means that we say, well, what's wrong about this is not

Timestamp 13 minutes the choices about when and how people can park, but the way in which those choices are being presented.

And it involves thinking through sketching and other tangible representations.

So perhaps as people were moving from a bad design to a good design, they might have had multiple ideas about how to do this. Many of them would have been bad, and then they would have ended up with the one that seems to be better than the rest.

And finally, the design addresses instrumental, technical, aesthetic, and ethical aspects throughout.

So what that means is that it's really trying to be effective, it's trying to address the pragmatics, pragmatics, the technical issues of is it possible? It often tries to be quite pretty, aesthetic, and very importantly, it needs to be

ethical. So we need to think about how will people use it and why. If the right design was something that was misleading, for example, and we've deliberately designed it so that people would think that they're allowed to park but would end up getting a parking ticket, the ethical aspect of that would be problematic. And, well, that's what many sites, for example, do in terms of dark patterns trying to force you to click through.

The idea is that if we're trying to do design, we should try and keep it as ethical as well.

So in this sense, we're already all interaction designers, because we continuously shape and create spaces in which we work, live and play.

My workspace might not look exactly like this one, yours probably doesn't either, but I'm sure you've deliberately moved things around to ensure that it supports you in whatever it is that you want to do.

So what I want to do now is to show you how we are already highly attuned to design aspects within the context of what we might normally do, our usual activities.

That is, we already, all of you, already think like designers without realizing it.

And the way how I want to do this is that there's this example of an architect, artist, who's built

Timestamp 15 minutes these objects that are called

the uncomfortable and I really invite you to visit the website but they're that are just really beautiful and exciting so let's have a look at two of the objects that Katarina developed now these objects are beautiful but also quite disconcerting you can see what they refer to pair of wellies and a fork but our minds automatically simulate how the interactions that are enabled by these objects would look like and how they would just go entirely wrong and it's clear that they would not lead to the goals that we might have. Wellies are about getting our feet to remain dry whereas these wellies kind of seem to do most of the work but the cross with sandals just like totally spots everything that the rest of the object is trying to do. Similarly for the fork, part of what makes fork a fork is the fact that we can hold it nicely and that it fits into our arm and that it allows us to have some leverage against the food. Now this is exactly the kind of thinking that designers will

Timestamp 16 minutes have for apps, for human AI collaborations too.

It's about thinking how does the object that you just created, how will that enable people to do the activities that they want. So you already think like designers. Let's have a look at this

example. This might be slightly less obvious for those of you who wear glasses you will automatically probably know what's wrong but for those of you don't it looks very pretty it's it's a beautiful pair of glasses it looks a bit like owl for me it reminds me of harry potter just because my daughter kept on watching that but the problem here is that if you imagine what it would feel like to put these glasses on your head then you'll realize that that one single point exactly where all of the weight of the glasses is being put is entirely wrong, right? That's what makes it uncomfortable. We will know that the interaction will not be nice. And our brains already help us see this and simulate all those interactions. That is interaction design. My favorite simplification though is that interaction design is really designed for people using technologies to undertake activities in contexts. Now what does that mean? So it's not just the interface or the artifact that we design, it's also about how people will use it, remember Douglas Glasses, and what they assume it is for, remember the well-ease. In other words, what the affordances that the object provides you with are. We will talk about that concept more in the next segment, but for now let's just

try thinking about what
is wrong with this design.
Why would it be dangerous?
doesn't seem to be
problematic, right? Canola
cooking spray probably means
it's oil. You can spray
it on the pan, like, it
might blow up if you put
it close to the flame,
but a lot of things do.

So, like, what is the
problem with this?

Now, imagine you
also now have this.

The issue is not the
individual objects. It's the
idea that we now have
two things that look very
similar, and if they were
standing next to each other,
next to a hob, you might
end up spraying fly insect
killer instead of your
oil, or vice versa, spray
oil all over the kitchen
to try and kill them all.

How about this one?

We might have a pan
that's trying to do
some education
lesson, but the moment
people start using it,
the problems arise.

So, this is what we
mean when we talk
about designing for
people using technologies
to undertake
activities in contexts,
where the interaction
design is really about not
just the object that
you're designing, but also
where it's going to be
used, how people will end
up using it, and what are
the other things in the
environment that you have
to be cautious of, as

and when you're thinking about your technologies. Timestamp 19 minutes So what I'd like to leave you with for this segment is really that I'd love for you to remember this idea of interaction design being a way of thinking about designing for people who are using technologies, which might be artifacts that might not necessarily be digital, to undertake specific activities in specific contexts. And what we're going to be talking about, so all of human AI interaction and user-centred design is about how do you build these technologies that are easy to learn, effective to use and providing enjoyable user experience. Now, as the final part of this segment, I would like to point you to a thoughtful but short blog post, which is part of your mandatory reading materials for this week, and which outline how these general design principles remain constant within the new Gen AI world too, but that also the specifics of how we think about doing design can change quite markedly. I suggest that you read this before moving to the next Timestamp 20 minutes section, and I'll look forward to seeing you there.

Timestamp 0 Alright, in this and the next segment we'll start talking about affordances which is an interesting theoretical and somewhat tricky concept that is

at the heart of many design guidelines and tricks that we'll be talking about in the next weeks.

A simple test of how successful the segment was for you will be whether this little image will start making sense to you when we're done.

So remember how when we talked about the objects from the Uncomfortable website, most, if not all of you could probably see straight away what these objects could or could not be used for.

Affordances are the key theoretical grounding for such process of understanding what is potential purpose of an object, and are the key idea for these natural interactions, especially if we're talking about physical objects. There is some relatively deep and somewhat confusing philosophy behind this notion, but for our purposes let's focus on the surface level characteristics.

So let's look at the definition.

What are affordances?

The term affordance refers to the perceived and actual properties of the thing, primarily those fundamental properties that determine just how the thing could be possibly used.

So what does this mean?

A first part of the definition, fundamental properties, is essentially the functionality that is enabled by the object.

So these special wellies afford walking in

water, although perhaps
not keeping your
drive, and a special
fork affords holding,
although perhaps not
particularly easy eating.

The second part of the
definition talks about the
perceived and actual
properties. What this means is
that the affordance refers
to not just the objects, but
to a relationship between
the object and the user.

So what does this relation
mean? In the context of
design, affordances were thought
of as directly, naturally
perceptible clues about
what the objects are for.

So when affordances
are taken advantage of,
the user knows what to
do just by looking. No
picture, label or
instruction is required.

Complex things may
require explanations,
but simple things
should not. When simple
things, such as doors,
as we will see soon,
need pictures, labels
or instructions,
the design has failed.

So when you think
about a button, when
we see a button we
kind of assume now
that it's something
that can be pushed.

If we see a switch, a
lever, we know that it can
be moved up and down or
whatever the axis is. And
if we see something that
looks like a knob, we
now kind of assume that
we'd be able to rotate it.

So, what happens

if affordances are
confusing, when we
think about the object
having some property
but it does not?

So in this case, if you
see a door and it has
a handle like this,
well, you're probably
going to hold the handle
and then pull, because
that's what the handles
are usually for. So
that makes all the sense.

But now think about
what happens from the
other side of this door.

It's nice and symmetric,
but it means that when
you come to the door
from the other side, you
will have a handle but
if you try and pull it's
probably not going to
do anything. So the
assumption is that on the
other side there will have
to be a little thing
that probably says push.

So what this suggests
is that what is
perceived as an affordance
might be misleading.

There might be situations
where seeing an affordance,
so seeing a handle and
being able to pull at a DAW
is good. It means it's
possible and it's perceptible.

So it's a perceptible
affordance and it's one that
we can do at other times
there's no handle you
don't want to pull that's
also good you you know that
that is not an interaction
that you can have with
that object the two problematic
states are if there is
perceptual information so

for example the the handle
that allows you to pull
but the door doesn't allow
you to pull or if it's
hidden so you have a door
that you don't know how to
open because you're supposed
to push and you haven't
realized that's the case.

So the idea is that we
have desired actions.

That's what we want to
enable. That's the interaction
that we're trying to
design for. And these
design actions cannot be
carried out if the object
does not afford it,
meaning I want to push, but
the door doesn't open that
way. And afforded actions
might not be carried
out if the user doesn't
perceive that these are
possible. So again, a door
might be very easy to be,
like, lifted up, but if
I don't know that a
lifting up is a possible
action with that door, I
will not be able to do it.

So in this example,
from this side, the handle
seems to be a perceptible
affordance. I mean, there's
additional information on
the side that guides us to
it as well, but we can pull.

On the other side, it
might be more problematic.

So just to finish up with
this boring door interlude,
if designers were in
control of the world,
this is how all doors
would look like.

Timestamp 5 minutes If you're supposed to push
at a door, it does not
have a handle, it just
has a little strip that

indicates this is where
you push. This side of the
door is where you push.
If you're pulling, you
have a handle, you hold
it, you pull. And if you're
supposed to turn, then
you probably have a knob.
as you go outside,
after you finish
watching this lecture,
wherever you are, go
and have a look at
a couple of doors.

It's very likely that many
of them will not follow
this pattern, and that you
might have had, probably
had that experience of
trying to walk through a door
and pulling at it or pushing
at it, because that's
what it seems to afford,
and then banging your face.

In effect, whenever the
functionalities are not
naturally obvious, you might
need to add more design
elements that can supplement
and provide cues to the user
as to how they can or
should interact with your
system. Now we've been
mostly talking about physical
objects so far and that's
just because they're quite
simple and understandable
and these affordances are
something that is quite clear
and close to the functionalities
you might want to do.

So you could see the
wellies and the affordances
that they have or you could
see the fork. The same ideas
apply with digital objects
too, and especially ones
that are powered by AI.

So the idea of desired
actions and afforded

actions plays out
with digital systems
too. So this little
video shows what seems
to be a relatively
simple interaction,
which is actually not that
simple, but it's one that
we've already been well
trained to do. So there's a
text box, we can probably
write there, things appear
underneath that get highlighted
as we mouse over, so
that's probably clickable,
and when we need to send
something to the AI, we're
getting the answers back.

So the desired actions here
are to search, potentially
to get some suggestions
and save some typing, as
well as be able to decide
when the query is complete
and I want to hand it over
to them to get answers.

The afforded actions
are also very close.

We can type, we can
select, we can send.

Timestamp 7 minutes Now this is a
different example.

One that is, again, using
the power of AI, but it's
much less clear, and I've
taken this from Adobe
websites, where it seems
that you're capable of
manipulating images in these
really complicated ways,
but at least for myself,
when I haven't been working
with the system, what
the affordances are, what
the desired actions even
would be that I'd like
to carry out, and what the
afforded actions are just
isn't as clear compared
to the previous example.

So what is the takeaway here? The whole idea of affordances is just a way of helping us think through how when we're designing user interfaces and experiences, we kind of automatically use queues that suggest actions that are possible by an interface element.

If we have a stack of cards, that might allow us to rearrange and shuffle. If we have an underlying text on a website, it suggests that it's a hyperlink and it can be clicked.

If we see something that looks like a welly, we kind of automatically assume that it will keep our feet dry.

But we're still figuring these out for AI, and that's what the rest of the course is also about.

So to end this section, I'll be pointing you again to a fairly lovely blog post by the same person who has started about affordances in the 80s, mostly for physical design at that point, and is now describing how the concepts can have purchased and be used within digital systems as well.

So please pause the video, go and have a read before you start the next segment, as I'll assume you will have at least kind of skimmed or read and ideally understood the main ideas of the text there.

Okay, welcome back. I'm assuming you've read the text so you know what

signifiers and affordances are,
but just to double-check
and define them together.

Affordances, as
we've talked about,
define what actions
could be possible.

And signifiers then specify
how people discover these
possibilities. So signifiers
are signs, perceptible
signals of what can be done.

So the pieces of affordances
that can be perceived by
people are signifiers, and
if these signifiers are
created by the designer, it's
called something that's
called of a social signifier.

So what does this mean? And
again, sorry, we're going
to stay with doors just because
doors are lovely, people
like them or not, but more
importantly, they're quite
simple and good examples
to show these constraints.

So when we look at this,
we kind of know what
we've talked about the handle
as being an affordance,
right? So it's something
you can hold and often
you might be able to
pull. Now the signifier
is suggesting and explaining
what to do, because
affordances, in this case,
seem to be confusing.

Let's have a look
at another example.

In this case, this is
a little more exciting,
right? So we have a
signifier with some

Timestamp 10 minutes icons showing what
is locked and what is
unlocked, because that
is not entirely obvious.

And we also have some

affordances, so there's a button for opening that seems to be just about the same size as a thumb would be, so that's an affordance. it's something that you probably can click. And then you have a sliding lock that will be used to really lock the coffee mug, and that's what the signifier suggests.

Let's have a look at a digital version of this. In digital versions, the affordances might be more hidden, right? Because they're digital. So in this case, the affordance is the fact that we can tap onto something. It's the button itself, the tap target that sets up a set space, an attach system, And it's not really the visual representation of the button, you could have an affordance, the ability to tap, even though there is no button there that would signify that that affordance is there.

Then we have the signifier, which is a visual Timestamp 11 minutes representation of the button, a light bulb, icon to tell you what's happening, whether you've tapped it, a label to tell you which button or which room the button relies on, and some sort of additional information about the bulb. So all of those are signifiers about the underlying affordances.

So to finish up these examples of the affordances and their relationships to signifiers, let's go back to the

chatGBT example that we've had before as part of the video. I won't label everything, but just to give you some ideas, the fact that we can type is a particular affordance.

We might have a little signifier that suggests that that button kind of sends, that's the little arrow.

we have another signifier that kind of highlights as you mouse over that you're able to pick and that this is something that will be reactive and so on.

So the whole idea is that this notion of affordances and signifiers allows you to think about what are the Timestamp 12 minutes capabilities and interactions that your technologies afford and then also how do you make these visible and available and understandable to your participants.

So again, affordances define what actions are possible, signify or specify how people discover this. I think that what I really like about this particular picture is that when you do look at a tree trunk, you kind of can see, it's a physical thing, you can see what that object will be used for. Because you've seen so many other objects that looked like it, and there's something about chairness, because it's the right height, it's the right width, hopefully it's not too sticky or too dirty, but you can imagine sitting on it.

Within the digital experiences, that's a little bit harder.

So the idea is that you have a bunch of pixels, and some of these pixels have some properties, such as that they're pushable, that you can do something, whereas others do not. So there is this idea that signifiers become really important within the digital environments, because the affordances often end up being exceedingly hidden. So the idea of kind of being able to use and build on what we'll call mental models, which is something we'll be talking about next week, to help people understand which parts of the screen are pushable and which parts of the screen you can select or scroll through and so on, all of these will be ways in which you apply that design thinking and that design sense and use signifiers in ways that people understand.

A lot of this thinking comes from a very famous and now very senior professor at UCSD called Dan Norman, and I'd really recommend you reading these books.

Obviously, they're not a mandatory part of reading, but if you're interested in design, or if you're interested in potentially taking this as either a career or as a research career, these two books are really, really exciting, and you will see all the professors and all, and most of the UX practitioners also know these two books quite well, because even though that they're reasonably old by now, The ideas, the ways in which

Don Norman's capable of talking about the principles of how human cognition works and why you work with design the way you do, those just remain true, even now.

And I'll just want to leave you with this one little example. So what are the affordances here on both the original and fixed version?

Have a think. What are the signifiers? How does this match with the three design goals that we've talked about in the first session? Just pause the video and think about it. Are you able to apply these concepts to this particular image?

Is it clear to you? If not, read more, go back to the videos, think about it. This is what most of the course will be about. There's some theory but now you have another example of a design problem.

How do you apply it here? What does it say? What can you tell about it? How would you redesign this?

Timestamp 0 Hi and welcome back. In this last segment for this week, I'll try to briefly introduce the processes of user-centred design, which you will then hear much more about in weeks four to six. I'll start with potentially slightly boring definitions, introduce a few more fun ways which people have previously tried to use to explain the process, often

in some form of what is called a double diamond, and then jump into an example of how a user-centred design looks like based on the research piece that we've done in our group over the last summer.

All right, let's get going.

As a quick reminder, what we've been talking about in the previous part of the lecture was all associated with what does it mean to think like a designer.

We've talked about the notion of affordances that are often visible for physical objects such as the uncomfortables here, as well as signifiers which take a higher importance in digital designs.

And also about how all design is actually situated in how, when, why, and for what people want to use what you created.

This is what makes the objects on the right so interesting.

They clearly nod at a particular use, but there is something about how these are designed that would either break the activity that's being supported, or just not work in the context where you want to use them.

We also talked about oversimplifying, but fairly useful way of defining interaction design goals as easy to learn, effective to use, and providing an enjoyable user experience.

At least one of these three goals is then what fails both for these

bad designs here as well
as for every single one
of the uncomfortable
ones. And I hopefully, I
think you should now be
able to identify and see
what these fails are.

All right, but so far
you might say, Oi, Peter,
all you've been telling
us is about that we
should know when stuff is
right, We should really
know when it's wrong,
but it's not really clear
how to get from the
latter to the former or
ideally decide the right
thing from the get-go.

So what should we do?
This process of getting
to the right results that
fits with what users need
is often encompassed by
what the literature calls
a user-centered design.

The latest ICO from 2019
defines user-centered design
as a approach to systems
design and development that
aims to make interactive
systems more usable by focusing
on the use of the system
and applying human factors
or ergonomics and usability
knowledge and techniques.

Well, that's a mouthful. The
Interaction Design Foundation
explains the same thing,
but in slightly different
words, focusing less on what
the aims are, more usable,
applying user factors and
so on, and emphasizing more
the process through which
these goals are achieved.

So the user-centred design
is an interactive design
process in which the
designer focuses on the users

and their needs in each phase of the design process.

So what this means is that in the UCD the design teams involve users throughout the design process through a range of or variety of research and design techniques to create highly usable and accessible products for people.

But what it really doesn't tell us is what are these variety of research and design techniques that we could use.

So let's jump into the overall models of the process.

What stages do these definitions expect you as a designer to go through? And I'll show a couple of examples of how the user-centred design process was conceptualized so you get an idea of what the common patterns are that seem to emerge across many of these ideas.

So in this process, which is the ISO 2019 human-centered design one, you start with planning the human-centered design process, because obviously that's important, but then you move to understanding and specifying the context of use. There are these magical user requirements, which is something we'll talk about as often being pretty hard to do most of the time, And then these arrows kind of walk you through, right? So you produce design solutions that meet those user requirements.

You evaluate those designs across or against those requirements. And then these kind of dashed arrows essentially talk about the possible failure points. You might have gotten the context and the requirement rights, right? You just messed up the design solution, so you go back to producing a different design solution. That doesn't fail your evaluation.

Or you might need to go all the way back and you say, well, we've clearly actually misunderstood what it is that we're trying to do.

We designed for the wrong context. This doesn't work. We need to kind of go back.

So overall, the key emphasis here is that this is not a waterfall model.

It's not something where you specify it up front, you have all the information, you decide what's going to happen, and then you just deploy.

It's much more similar to what might be happening in Agile. So a lot of iteration, a lot of assumption

Timestamp 5 minutes that you never have enough information at the start, and that it's through the design and through the process of building things that

you gain the information you need so that you can start making good choices.

From a 10,000 feet level, the process of design can also be seen as moving from a place where we don't really know and there could be a solution, and in fact

we probably can't even articulate the problem that's supposed to be solved properly, to a place where we only know what, not only know what the problem is, but we've already designed and tested the solutions and now, you know, they're out there, whether it's in your research thesis or out there in in commercial sphere. The double diamond model should not be seen as a linear process and we'll keep on seeing this kind of notion of diverging and converging and diverging and converging at least twice repeatedly. So it's not a linear process where you go from A to B and you kind of never stop but it's it's really supposed to be a map of the types of design thinking or the Timestamp 6 minutes areas of interest that you will likely visit along the way although you'll be cycling throughout and kind of coming back and forth, just like within the ISO.

So a couple of the main points that I want to highlight here are the big difference between doing the right things and doing things right.

What this is trying to say is that very often in design, you do not know what is it that you should even be building.

So often the first part of this diverge and converge system is really just about you getting the right understanding of what the problem is, what

is it that the users need,
what is their activity,
what is the context,
what do we need to know
about it, and what should
the system even be doing.

So this notion of a design
brief is a really critical
document that actually ends
up being a key contribution
for much of the work. So
this idea that you might
spend more than half of
your process, 60-70% of
your time, just even figuring
out what is it that the
design should do that is
entirely normal and natural
so the other piece that
we want to focus on is if
you move from doing the
right things and you move
to doing things right
again you will be expecting
that even though you've
had the design brief and
you know what you're
trying to do you will again
start ideating you'll
start having many ideas
and many possibilities
that you need to explore
before you start converging
down to the right solution.

So these diverge and
converge models are
something that very often
is seen as defining design.

And again, it's this idea
of thinking through doing
and never knowing all the
information you need to
have. So there's this
phrase where if we knew
exactly what we want to do,
you wouldn't need designers.

The design is about
understanding and mapping
some sort of difficult,
hard, tangly, wrangly problem

that doesn't necessarily
have a clean solution.
Alright, and the last piece
that I want to highlight
is that there's really
an expected iteration,
certainly within the
diamonds, so as you're
diverging and converging
there's a lot of iteration
between what you believe
the ideas might be, how
do you define your
research areas and methods,
you know, what do you do
as primary research? You
might find insights that
then lead to you having to
redefine an area and start
doing more primary research
and so on. As well as then
potentially across the
diamonds. So after you've
defined your brief and
you have some ideas and
you're trying to start to
deliver, so all the way
potentially even back here where
it says build a trade and
repeat, you might realize
that actually by the way
in which people started
using your system that you've
misunderstood something
really important and it
might kick you not entirely
back you don't forget
all the work that you've
done but there will be
phases of the first diamond
where you realize that you
don't know what the right
things are yet and you'll
have to go back so it is
confusing it's iterative and
it's it's certainly something
that's quite fun so to
illustrate the differences
between what does it really
mean to find a good design

brief and how might we do it with users. So there's this idea of we're talking about user-centered design. What are the users' needs? What are their hopes? What are their dreams? How will they use systems? But those systems don't exist yet. So there's this famous quote that probably has never been said. But if you think about Henry Ford as the person who designed the first car, he's told to be saying that if I had asked people what they wanted, they would have said faster horses, right? And this is all always or often presented as the idea of you have this tension of really trying to understand what people's needs are, but people usually don't know what the solutions to those needs would be.

So the follow-up to this is, but what Ford did notice was that when he asked what people wanted, the main thing they said was that they wanted a faster horse who doesn't get tired, rather than more comfortable.

I think this is what you'll be hearing much more about in weeks four, probably, and five, but this idea of being able to go to users and understand what they need, even though they might not know what the solutions to their needs are, is really critical and interesting tension within the Sun.

So just very quickly now, another way of simplifying the design process, especially around involving users'

needs, comes from the Stanford Design School and IDEO.

It's positioned again as a series of stages, but you'll see all these little arrows that indicate that in real projects you'll be flying back and forth between these.

And we've started to talk about little bit of some of these tools, about what is Timestamp 11 minutes it to empathize, what is it to define, but the rest of the semester will kind of be going through the individual stages, especially in weeks four to six, and you'll understand more about this.

So just to close off this section, I'm leaving a couple of slides here from IDEO, which is one of the most famous design studios on the planet, and I'll let you read these if you're interested, but this first slide just outlines the types of questions that designers will ask in different parts of the process and notice again how does diverge and converge models are happening twice just like in the diamond model right so all of these are restatings of this idea that seems to be really common where we need to understand what is it that we're even trying to design and then try to design it properly but in both cases we have to diverge to understand what the range of ideas are And then only then we might start

converging on what good solutions would look like.

And the second slide is one that shows the seven key lessons or the approaches that IDEO works. And

I really like this because it's clearly aligned

with the way that we've ended up doing design in my lab. But it also helps

us articulate how design differs from other science areas. For example,

if we're trying to introduce these ideas of design thinking or innovation

into clinical teams at Harvard, Cambridge, Oxford, various universities.

So if you have a few moments, just pause the video, have a read through.

You'll see that each of these is a way of thinking.

It's a way of approaching the world that is sometimes similar to what an engineer might do automatically, but often it is not. and it's it's a really interesting way of knowing and seeing and engaging with problems.

So much of what we've talked about now has been mostly focused on the traditional design processes as these still encompass a large proportion of

what you might do. But obviously we've been talking about these two different types of AI and how this will differ and impact these design processes.

So, to hint at the differences, I just wanted to leave you with this beautiful image that comes from a

relatively famous paper now,
which is about two years
old, that kind of indicates
the difference between
traditional design and
designing with these stochastic
amorphous systems that kind
of seem to wave and shape
underneath your hands.
Now, the top is a really
famous image it's called
the design squiggle it's
very often used to show how
much of a mess design is
and how you go from one side
to the other and how it's
really kind of messy and
all over the place at the
start but then over time
the amplitudes kind of start
leveling out and then you
have a clarity and focus
and that's when you've
actually started you know
prototyping and designing
the real system you're past
prototypes but you're now
you know what you're trying
to do you know what the
design is and now it's about
engineering and execution
and marketing and all of
these really important areas
but from a design perspective
you know now and underneath
you can see how that
same squiggle has been kind
of adapted into a never
-ending area which hopefully
might not be exactly the
case now it is two years
old but it indicates how
the generative AI AI
approaches might change what it
even means to design and
how we might deal with this.
So what I want to leave
you with today is this
beautiful blog by someone
who's a general manager

at Xbox and who writes about this idea of treating AI as design material, which is something that comes from art and design as making things pretty, where the idea is that each medium, so whether it's clay or wood or stone, if you're a sculptor, each of that medium has a life of its own and it kind of resists what you want to do. It might splinter. And the role of the designer is to really understand the quirks and the ways in which the medium kind of works in ways that you don't expect. So what this blog talks about is like clay resists, wood splinters and ai misinterprets it's it's something we need to learn to deal with and it's that material friction where good design happens i'd really encourage you to read this um it's not that this is going to be anywhere on the on the final test um it doesn't have a particular concept in it that um would be a theory that you need to understand that you know, AI does X, Y, and Z, but I think it's a really interesting way of getting your head around what does it mean to design and what does it mean to design with things like AI that we don't understand yet fully, we don't know how it works as designers, because it keeps on changing underneath our hands and we haven't quite gotten the rules of thumb or best practices down just year.

So please give this a read.
Alright, so to wrap up
with an example, I wanted
to do a quick run
-through of this work that
we've done with lovely
people across Cambridge
and Stanford and University
of Washington and
University of Michigan
last year over the summer.
So this is a research paper
that I've been pulling
pieces out of. It's also
something that hopefully will
be able to show you some of
the steps of the user-centred
design process that we were
talking about on something
that's ended up being
surprisingly interesting and
maybe will become reasonably
influential over time.
So it's something that's,
as I said, emerged from a
sprint over the summer where
we were trying to understand
how could we collect
information from young people
that would allow them to
articulate their difficult
experiences with social media
and the reason was that we
wanted to build a mental
health intervention or
cognitive reappraisal intervention
that would support young
people as they engage with
social media and potentially
struggle. So we're working
with James Gross who's
essentially the person who's
built up the whole field
of emotional regulation and
he's based at Stanford as
well, Sammy Orban, who's
focused very much on social
media and policy and social
media harms at Cambridge.
But within those 12 weeks

of inception, we've moved from not having anything and just having that idea to something that we built that already was shown to work in a randomised controlled trial and had interest from a range of researchers at leading institutions, you know, you can read some of the names, across domains, including basic psychology and clinical and human-computer interaction and implementation science and so on, as well as seeing this work or the tool being brought into a range of application areas, from maybe data gathering to starting to think about interventions.

So how did this happen? What was the process that underpinned this 12-week sprint from nothing to now we have a tool that a lot of people are interested in using?

I'll break it down into the design, according to the design thinking phases, just to give you a sense. So the first one's talking about understanding the user's needs and problems.

So what is it that we're even trying to solve?

So in our case, the users were actually these research groups, right?

So they needed to get this Timestamp 19 minutes information from hundreds of young people in ways that young people would be willing to share and would be able to articulate these experiences.

So the broader idea was that we actually realized that we don't have tools, we

don't have ways of collecting these narratives, these ideas, these stories from participants at scale without excessive participant burden or research costs. And what this means is that your traditional models are to interview people over Zoom or phone and as you might have noticed if anyone has ever asked you to do a research study, most young people just don't want to participate, it's not interesting and also it's just really hard to arrange that hour in your day where you fit with what the interviewer can do and what you can do and then often you don't get people showing up and so on. So we just had no way of collecting this data at scale. So that was our understanding and And what we've tried to really do was to think about, well, why don't people engage and what is it about the process that is hard? So we started thinking about, well, what would we need? We would ideally need, not very much, we don't need long stories from participants. We need stories that are relatively focused and succinct, but they need to answer all the pieces of knowledge that we need. So this is an example of one of the stories that we've gotten as the output. But if you had an open text form that just asked you, you know, tell me about your

social media experiences,
getting young people to
consistently write stories
like this in this format
seemed pretty hard, and
the experience suggested
that that wouldn't happen.

So our design challenge,
if you remember thinking
about design briefs,
Our design challenge was
that we're really trying
to develop what we would
call a remote narrative
elicitation method. So
what that means is a way
to help people articulate
Timestamp 21 minutes stories. That has the
following characteristics.

It provides enough
open-endedness because
we didn't want to
turn it into a survey.

We really needed
to collect stories.

It provides enough consistency
across participants.

So we wanted to ensure and
enable the participants to,
you know, not tell us too
much or not tell us too
little, really. And also just
focus on getting the right
answers that we needed and
not too much, again, just
to balance the burden that
we're placing on people.

And something that reduces
the cognitive burden, which
ideally will then mean
that it also increases the
willingness to engage and
also the ability to engage.

so i'm conscious that
we have over an hour of
recorded material by now
um anyway so i'm really
running through this
very quickly we might
talk a little bit more

about this um during the
large group tutorial or
lesson that we have um
together but the idea was
that if we have defined
Timestamp 22 minutes this this way how did
we start thinking about
well what the solutions
would look like right
so now we're moving
into the second diamond.

I've not shown you much
of the process, but
we've ended up in a place
where we now know what
our target is. We're
trying to build this remote
elicitation method
and it has particular
characteristics. So what
could the solutions be?

Our idea was that we
started thinking about, if I
asked someone to craft
these four to six sentences,
long descriptions, what
is it that they need to
be able to do? So what
is it that we're actually
asking them to do cognitively?

We're asking them to
know what to focus on in
that challenging situation,
which might or might not
be easy. If you spend
your life thinking about
all the things that social
media kind of worry you
or make you annoyed about,
well, it'll be an easy
story for you to tell. But
if you don't, if it's
just part of your life, it
might be harder to know how
to answer that question.

Timestamp 23 minutes We then wanted them to
know how much abstraction
they need to do. Is
it a word? Is it a
sentence? Is it a two

-page essay? What is the level of abstraction that we need for each of the aspects that we need to hear about?

Third point, I think, is the crucial one.

Yet people needed to both be able and willing to articulate a story that was coherent and that is kind of fitting into the format that we needed.

And finally, we needed to ensure that even if you write a story, it doesn't mean that it actually captures what your experience was or what you wanted to do.

So from ours, as we were ideating, we were thinking, well, okay, this is actually what we're trying to get people to do. So this became part of a design brave, but it happened as a way of thinking about what the solution would be. What is it that we could support for each of these key steps along this trajectory here?

Timestamp 24 minutes The underlying idea, and I

won't go into this, there's about 30 to 40 years of research in psychology on vignettes. So these vignettes are these relatively short stories, similar to the one that you've seen. But usually the way how they've been used was to have researchers interview a bunch of people and then make this story as if it was coming from a participant and then sharing out the same story with dozens of other research participants as a way of engaging them with the same content. So the idea was

that researchers were using vignettes as storytelling device, right? It was a way of saying, well, imagine you're in this situation that another student has done this and this and this and this as their part of preparation for an exam, how would you approach their situation if you were in their shoes, right? That's how vignettes have been used.

And what we said was, well, it's interesting.

Now that we have generative AI, can we enable participants to make these vignettes so that we use that same idea, but swap it around, where the participants are using the same technique and we enable storytelling from participants to researchers?

So, again, jumping ahead really fast, but based on these ideas, and based on the way in which those ideas then shaped the brief that we

were trying to build, we started prototyping and thinking about, well, how could we fix this?

Okay, so I'll jump ahead and not really talk too much about the prototyping, but really describe the final product. What did we end up doing?

The underlying system ended up being incredibly simple. It was just three sequential calls to an LLM that had been scoped in interesting ways, and controlled with a traditional state-based finite machine.

So very, very simple.

What did it end up looking like for the participants?

So there are kind of three parts, some of which are seen by participants, some of which aren't. So as a participant, you would get this conversation, so you would be asked a couple of questions, and you'll see that those have been more or less pre-specified, but the LLMs allow you to embed those questions in interesting ways. You would answer potentially with just a couple of words, you didn't have to even use the full sentences.

Then the second stage would extract your answers.

So that's if you think about the abstraction.

It's about being able to extract the right level of information that people have said and do that within their actual words. And then we used another piece which combined the extraction into a story. And the idea here is that if we have the right questions, that's what creates the template for a vignette, template for a story. and even if you then have these relatively small skeleton hunters you can use these as the core pieces of the story and then the other one can just finish the connective tissue. So it's the same idea of if I'm asking you to create a fairy tale for example I will say well who's the hero or heroine where do they come from what are they trying to do are they trying

to get married or Are they
trying to fight an enemy?

Who the enemy is? What
is the most problematic
thing that happens to
them? How do they resolve
it? And how does the story
end? If I have answers
to these kind of
individual small questions,
I can create something that
will look like a fairytale.

So this is exactly the
same idea we used here.

So if we step through
this, there's some notion
of we're delegating
cognition and scaffolding
and thinking about how
to design things in
ways that make it
simpler for participants.

But again, I won't go into
this in detail in this
video. But the first idea
was, well, people might not
know what to focus on. So
we're going to provide this
template, right? In the
same way as if you use the
fairytale story again,
it's easier to answer these
questions. Who's your hero?
where did he come from? What
is their goal? What's the
worst thing that will happen
to them? What's the biggest
risk that they have to
take? How do they resolve it?
And so on. These questions
are something you can
answer kind of one by one.

And you've built
the story, but you
don't have to think
about it. So that
same idea is what
we've done here.

The specific questions
in the case of

social media were,
well, what happened
specifically? What was
set, posted or done?

That seems easy for people
to answer. Once they've
answered it, well, what's
the context? What else do
we need to know about that
situation? How did that
make you feel? How did you
react? What was the worst
part of the situation? And
our idea was, well, if we
Timestamp 29 minutes have answers to all these,
we kind of have all we
need and we just need to
craft it into this little
narrative, into this story
that people can react to.

So we had a single component
that just went and asked
those questions with a little
bit of scaffolding around
them. And it seemed easy
for participants to answer.

So what was the next step?
We now have a dialogue
that has answers. We're
talking young people. Those
answers probably won't be
full sentences. So we want to
extract the exact words,
kind of the core of what
their answer was to each of
those questions. And again,
that's relatively simple.

We can trust the LLMs to
do that reasonably
consistently, and we can also
control the level of detail,
so this abstraction of how
much information we need
to do, we need to collect.

The third piece was that
now we have that data
extracted, we want to craft
it into a story, right? So
it's the same idea of I have
the four or five elements

of what makes a fairy tale, now I need to build a fairy tale around it. I could build a big one, I could build a little one, but in our case we were really trying to take answers to those four questions, put them one after another, and then ask the LLM to craft and add anything they needed into just making it sound like a story. But we knew that it could be a story because of the way how we've crafted those questions and that template. What we also wanted to do, and this is a design choice, we didn't want the participant to see just one story that were created out of their data, because that would mean that they're more likely to accept it. So we wanted to show them three different stories that were all created with a different emotional tone. So the idea of these personas that you can kind of see here was that you, as a participant, would get the three stories, each of which contains exactly your words, but the tone of them might be slightly different. So it could be coming from a teacher, or it could be coming from a friend or it could be coming from someone who's slightly younger than you. And then the final piece was we asked participants to review these three stories and to select one that they like and then adapt it. So the idea was, hey,

here are three things that are more or less probably close to what you wanted to say, but they're not going to be exact. Pick the one that's closest to what you're trying to say and then adapt it so it's actually really what you're trying to do something. And again, that seemed relatively simple for participants and also relatively simple to do with LLMS.

So the idea behind all this was that it was very easy to adapt because all of what you need to change, if you want to not ask about social media, but you want to ask about their favorite fairy tale, you just change the template and you change the personas.

Everything else in the system stays exactly the same.

So the idea now is that all you need is this little config that allows you to specify what are the questions and all of a sudden the tool can not just be used for social media context but it can be used for many other situations.

So we now built something, we were thinking how clever we are, this is amazing, we have a prototype, it seems to work for us.

But we needed to test it. So what did we do?

We ran a mixed methods experimental study. So what we've essentially tried to do was we got 250 young people on a platform called Prolific

to engage with two types of story generations each. And we'll talk a little bit more about this in week seven, I believe, when we talk about evaluation. But the general idea was that we really wanted to see and contrast what happens when we ask a young person to do the same thing with an open text survey, so just the traditional model that you might get in your exam

Timestamp 33 minutes or, sorry, in your evaluation of courses, please provide more details about what you liked or disliked about this course, so that kind of open text survey, or the system that we've designed for people here. And we did it in a reasonably clever way. This was run with Stanford. So we did it in a way where we could both do implicit comparisons. So kind of between subjects where people have seen either only the bots or only the forum, and we could compare it there. But we could also compare within subjects. So both each of the people have ended up experiencing both of those things in the end, so we could ask them which one they preferred. So we had both this kind of independent measures of how useful or unuseful these things were, but then also we had preferences metrics that were interesting. So what did we find out? It turned out that whatever we measured, the system we've design was just so much better than the open text form. I won't go through all

the details, feel free to stop the video if you're interested. But what was really interesting for us was really this notion of people were talking about how it was more helpful for articulating their experience, that it might have been less difficult to respond to, as well as some of the other aspects that I'm pointing out here.

We also, since we've collected an actual dataset, this is again done with Cambridge, where we have now 900 stories of young people across US and UK about the biggest challenges that they face. The reason why I'm including this slide is also because we then asked young people to tell us about their value. And remember, these were young people who were essentially hired to go and do this three-minute text survey, so Timestamp 35 minutes they really shouldn't have found it helpful. But what's interesting, if you pause the video, you can look at how big percentages of people were reporting that the experience was not just a thing they did for money, but it seemed to have been interesting and helpful, and there's some interesting thoughts behind why that might be the case and there's a series of studies that we can talk about um in in the actual lecture that why we think this is the case so just just to wrap up the the broader implications of this were that we started with a problem

that was our own little
problem we needed to
understand what young people
think about social media
but then we've ended up
building a system that
addresses is a much bigger
question of these collecting
rich qualitative narratives
at scale, this idea of
how do you make a good
design that would reduce
the burden for participants.
And we've made it into
something that's quite simple
to adapt. And we now have
a lot of interest in the
data collection and
interventions. This is the slide
from last year. So the
numbers would, sorry, the
little logos would be much
more multiplied by now. But
even within those 12 weeks,
we've had so much interest
from a lot of people,
which often suggests in
design that you've actually
had something that's
important where there's
need and where people need a
solution and they don't have
a good solution right now.
I wanted to pull out
this blog one more time,
because I think the example
that we've just gone
through is a good
illustration of what it
means to work with AI as
design materials. And if
you were to read the
underlying paper, or I'm
happy to talk about it
when we meet in person,
again, the idea is that
we've really worked with AI
as a material. We've
deliberately made choices about
the ways in which we're

willing to engage with it. How do we design around the quirks that it has, but also how do we design around the really fascinating properties that it allows us to do now. And there's a reason why no one's done that kind of tool ever before, which is, it just wasn't possible a year or two ago. So I think it illustrates both this particular, the ideas from this particular blog, but also the excitement and opportunities that lie around human AI collaboration space, which is perhaps why you're here with us in this course.

So as the very, very last part, I just wanted to highlight this article and TED Talk that you have available on Keats and that I'd really encourage you to read and watch. because I think it's coming from someone who has decades of experience in designing.

And I think it really illustrates and shows how much further you can go when thinking about design than just building yet another app that does exactly the same thing as all the other apps have done before.

But now it does it with a slightly different shade of gray or blue or pink or whichever color you like.

I really like both the article and the TED Talk. it's probably not going to take you that long, and I hope that you'll find it really interesting. So with this, and as the end of the

first week, thank you so
much for being with us,
hoping that this is something
that seems to be relevant
and interesting to
what you want to do in
future, or what you want to
learn, and let me know if
you have any questions.

This will be a relatively
small class, I believe
for the first year so
I'm really excited to
meeting you all very soon
all right see you soon bye